Geothermal Energy

Chapter 8

A person purchases land to build a cabin. But there are no utility poles or pipelines nearby to carry electricity, natural gas, or water to the cabin. Instead, the builder uses solar panels to produce electricity from solar energy. For heating, they use a geothermal heat pump. The pump moves liquid through pipes that run deep into the ground and back to the home. Heat from underground, which is always about fifty degrees Fahrenheit, warms the liquid. The liquid is then pumped into a unit that collects heat from the liquid. The unit distributes water, which is now much warmer than the air outside, throughout the house.

Big Question

What is geothermal energy, and how do people use it?



This site in Iceland is known for its warm lake and geothermal power plant. The same thermal energy that allows people to bathe in this cool environment provides heat for homes and businesses.

This is one way of using **geothermal energy**, or the thermal energy that is underground. There are many costs and benefits of using geothermal energy.

Vocabulary

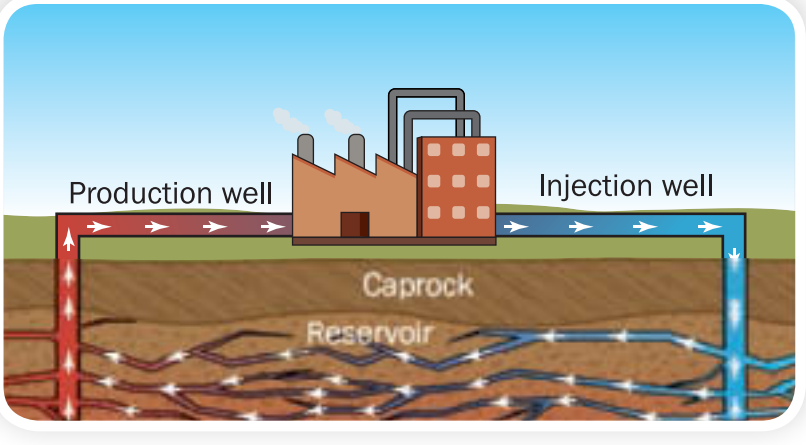
geothermal energy, n. energy produced by heat that is transferred from Earth's interior

Requirements for Using Geothermal Energy

Another kind of geothermal energy is available when very hot parts of Earth's mantle are close to the crust. This thermal energy produces hot springs and geysers. Such heat from the mantle can only be reached in certain locations, but it can have many benefits.

This kind of geothermal energy can be harnessed to heat homes, businesses, and other structures. Geothermal heat is used to create steam. The steam or hot water is passed through pipes that heat water in a boiler. This water can pass through pipes in a home and release heat as steam. It can also be used to cook with or bathe in.

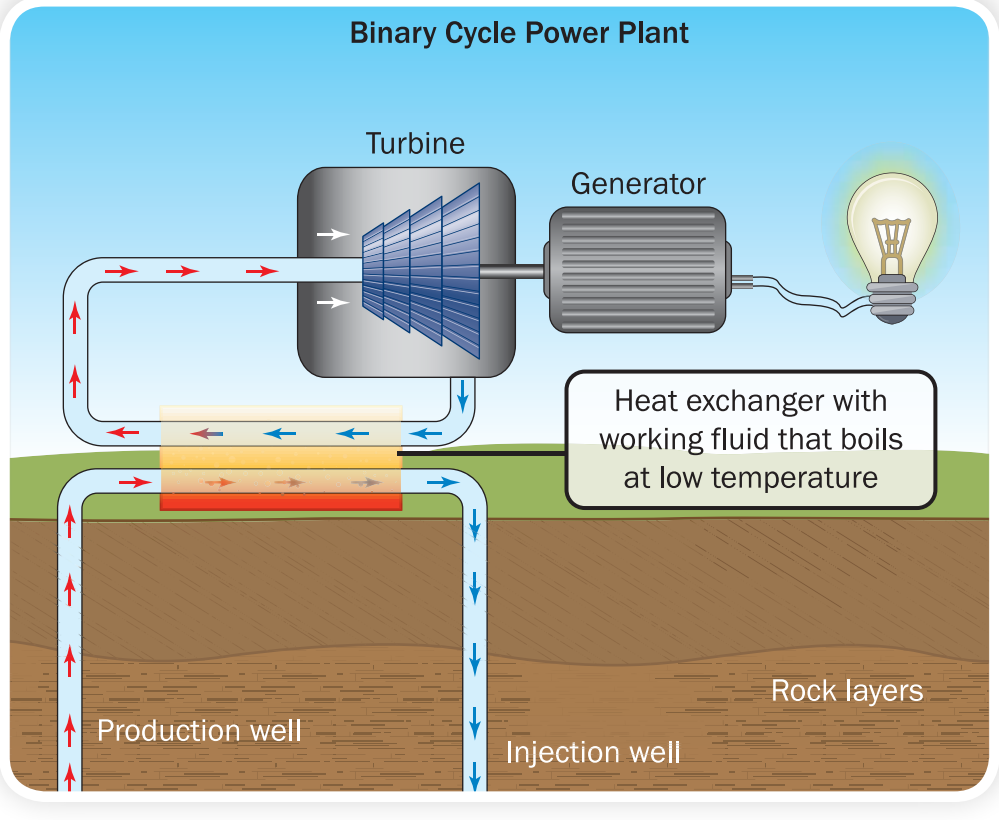
You cannot just dig a hole in the ground and find a source of steam or hot water. Most of these sources of steam or hot water are in areas where Earth's crust is active, such as volcanic areas. A well is drilled through the first layer of rock into the reservoir where steam or hot water is. After the geothermal power plant converts the thermal energy into electricity, the cooled water is pumped back into the reservoir through a different well. This recycles the water.



Geothermal energy is considered clean because it produces very few greenhouse gas emissions. It also does not take anything from Earth's crust other than heat.

Types of Geothermal Energy Use

Another type of geothermal energy can be extracted from old oil fields. Oil fields that no longer have petroleum but are full of wastewater can be used for geothermal energy. The fluid is not clean or hot enough to directly drive a turbine. Instead, it is passed into an apparatus called a heat exchanger. This transfers the heat from the liquid to another fluid that boils at a lower temperature than water. Therefore, it turns into gas without needing to be as hot and becomes vapor to drive a turbine.



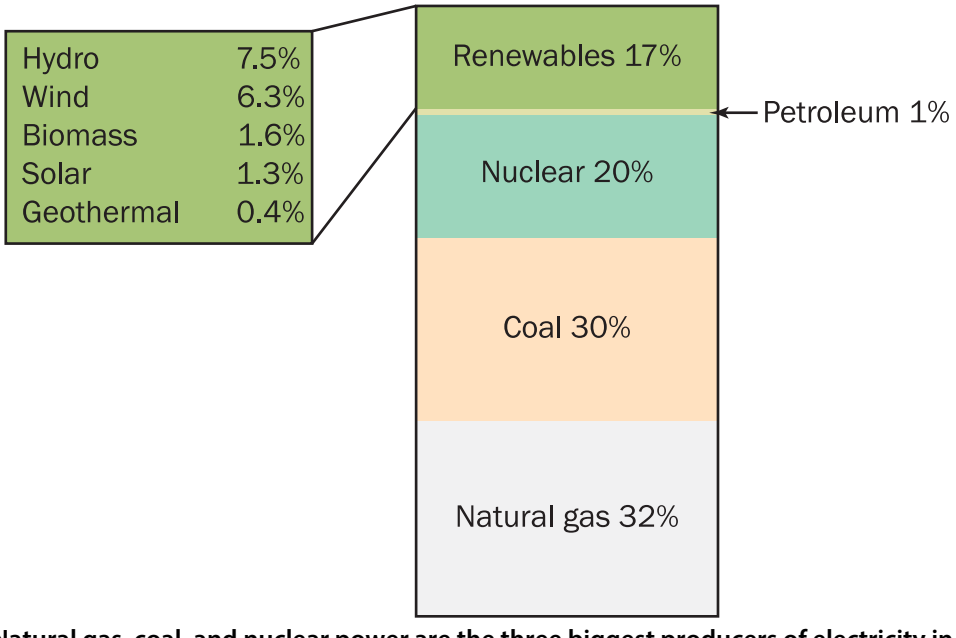
A low-temperature geothermal power plant uses cooler fluids. The red arrows show the flow of liquids or gases that are warmer and the blue arrows show when the materials are cooler.

The Costs of Geothermal Energy

Geothermal energy is a clean way of meeting our energy needs. However, it cannot be done on a large scale without a big investment of money. A safe site where geothermal fluid can be tapped must also be found. Iceland and the Philippines are two countries that meet a lot of their energy needs using geothermal energy. But both have access to geothermal reservoirs thanks to their locations. There are many volcanoes in the area. The United States has far fewer volcanic areas. But it still uses geothermal energy to generate electricity in some places.

Drilling the wells for a geothermal power plant can cause natural gas to escape. Unburned methane is a powerful greenhouse gas. Any opening in Earth's crust that allows trapped methane to escape could cancel the positive effects of operating a geothermal power plant.

Sources of U.S. Electricity Generation, 2017



Natural gas, coal, and nuclear power are the three biggest producers of electricity in the United States. Renewable resources account for seventeen percent in total, with geothermal providing just 0.4 percent.